Discretization, Linearization and Projection of the Navier-Stokes equations

to obtain a control-oriented distributed state-space model
DRAFT

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1. Introduction

This report presents the linearization and Helmotz-Leray projection of Navier-Stokes equations describing, in a simplified way, the aerodynamics of a wind park in 2-D. The Helmotz-Leray projection chooses the pressure field such that the continuity equation holds, and transforms the system of differential algebraic equations to a system of ordinary differential equations. Through these steps, we find an LTI dynamical model in the state-space form, which may potentially be used for control of the wind flow in the wind park and the power produced by each turbine. To generate the simulation results and see the different steps described here implemented in MATLAB, see the function `projection_dae2ode.m` in WindFarmSimulator version 2.2.2, to be found on the DCSC DataDriven disk in the Software directory.

2. Spatially Discretized Navier-Stokes

In [1], [2] we find the different steps of the spatial discretization of the 2-D Navier-Stokes equations. If we assume incompressibility, i.e. constant air density ρ, and use hybrid discretization in the spatial domain on the staggered grid (see Figure 1), we obtain the following relations for the spatial and temporal variation of the pressure p, and the wind speed vector $\mathbf{u} = [u, v]$, with x-direction component u and y-direction component v:

$$\frac{\partial}{\partial t}\begin{bmatrix} u_{i,J} \\ v_{I,j} \end{bmatrix} = \begin{bmatrix} \frac{1}{\rho V_{i,J}}\left(a_x\left(\mathbf{u}\right) - a_{xc}p_{I,J} + a_{xc}p_{I-1,J} + S_{x,i,J}V_{i,J}\right) \\ \frac{1}{\rho V_{I,j}}\left(a_y\left(\mathbf{u}\right) - a_{yc}p_{I,J} + a_{yc}p_{I,J-1} + S_{y,I,j}V_{I,j}\right) \end{bmatrix}, \tag{1a}$$

$$\text{s.t.}: \quad a_{x,c}u_{i+1,J} - a_{x,c}u_{i,J} + a_{y,c}v_{I,j+1} - a_{y,c}v_{I,j} = 0, \tag{1b}$$

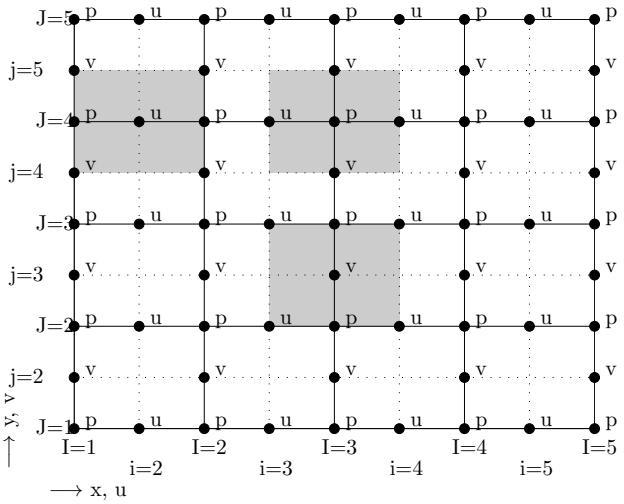


Figure 1: The staggered grid, from [2]. Gray squares indicate control volumes. On the left, we see the control volume $V_{i,J}$ used in discretizing the x-momentum equations, on the top-right the control volume $V_{I,J}$ used in discretizing the continuity equation, and the bottom right control volume $V_{I,j}$ is used in discretizing the y-momentum equations.

with:

$$\begin{aligned} a_x(\mathbf{u}) &= -a_{x,P}u_{i,J} + a_{x,W}u_{i-1,J} + a_{x,E}u_{i+1,J} + a_{x,N}u_{i,J+1} + a_{x,S}u_{i,J-1}, \\ a_y(\mathbf{u}) &= -a_{y,P}v_{I,j} + a_{y,W}v_{I-1,j} + a_{x,E}v_{I+1,j} + a_{y,N}v_{I,j+1} + a_{y,S}v_{I,j-1}, \end{aligned} \tag{2}$$

and where S_x, S_y are source terms (averaged over the grid cell) which in the wind park model are used to describe the thrust forces of the different wind turbines. We refer to the dynamic system (1a) as the momentum equations, and to algebraic equation (1b) as the continuity equation or the divergence-free constraint. For simplicity we omit a turbulence model (see [3] to see how to incorporate the -highly simplified- mixing length turbulence model).

The coefficients $a_{x,c}$, $a_{y,c}$ in the pressure terms and the continuity equation, are given by the sizes of the cells:

$$a_{x,c} = y_{I,j+1} - y_{I,j}, \quad a_{y,c} = x_{i+1,J} - x_{i,J}.$$

The other coefficients $a_{\bullet,\bullet}$ are a function of the approximate flows across the cell faces. See [1] for the derivation of the above coefficients. To be able to explain the linearization

of the model in section 4, we show them here:

$$\begin{aligned}
&F_{x,w}\left(u_{i-1,J},u_{i,J}\right)=\rho\left(y_{I-1,j+1}-y_{I-1,j}\right)\frac{u_{i-1,J}+u_{i,J}}{2},\\
&F_{x,s}\left(v_{I-1,j},v_{I,j}\right)=\rho\left(x_{I,j}-x_{I-1,j}\right)\frac{v_{I-1,j}+v_{I,j}}{2},\\
&F_{x,n}\left(v_{I-1,j+1},v_{I,j+1}\right)=\rho\left(x_{I,j+1}-x_{I-1,j+1}\right)\frac{v_{I-1,j+1}+v_{I,j+1}}{2},\\
&F_{x,e}\left(u_{i,J},u_{i+1,J}\right)=\rho\left(y_{I,j+1}-y_{I,j}\right)\frac{u_{i,J}+u_{i+1,J}}{2},
\end{aligned}\tag{3}$$

$$\begin{aligned}
&F_{y,w}\left(u_{i,J},u_{i,J-1}\right)=\rho\left(y_{i,J}-y_{i,J-1}\right)\frac{u_{i,J}+u_{i,J-1}}{2},\\
&F_{y,s}\left(v_{I,j},v_{I,j+1}\right)=\rho\left(x_{i+1,J-1}-x_{i,J-1}\right)\frac{v_{I,j}+v_{I,j+1}}{2},\\
&F_{y,n}\left(v_{I,j-1},v_{I,j}\right)=\rho\left(x_{i+1,J}-x_{i,J}\right)\frac{v_{I,j-1}+v_{I,j}}{2},\\
&F_{y,e}\left(u_{i+1,J},u_{i+1,J-1}\right)=\rho\left(y_{i+1,J}-y_{i+1,J-1}\right)\frac{u_{i+1,J}+u_{i+1,J-1}}{2},
\end{aligned}\tag{4}$$

$$\begin{aligned}
&a_{x,W}\left(u_{i-1,J},u_{i,J}\right)=\max\left(F_{x,W}\left(u_{i-1,J},u_{i,J}\right),D_{x,w}+\frac{1}{2}F_{x,W}\left(u_{i-1,J},u_{i,J}\right),0\right),\\
&a_{x,S}\left(v_{I-1,j},v_{I,j}\right)=\max\left(F_{x,S}\left(v_{I-1,j},v_{I,j}\right),D_{x,s}+\frac{1}{2}F_{x,S}\left(v_{I-1,j},v_{I,j}\right),0\right),\\
&a_{x,N}\left(v_{I-1,j+1},v_{I,j+1}\right)=\max\left(-F_{x,N}\left(v_{I-1,j+1},v_{I,j+1}\right),D_{x,n}-\frac{1}{2}F_{x,N}\left(v_{I-1,j+1},v_{I,j+1}\right),0\right),\\
&a_{x,E}\left(u_{i,J},u_{i+1,J}\right)=\max\left(-F_{x,E}\left(u_{i,J},u_{i+1,J}\right),D_{x,s}-\frac{1}{2}F_{x,E}\left(u_{i,J},u_{i+1,J}\right),0\right),
\end{aligned}\tag{5}$$

$$\begin{aligned}
&a_{x,P}\left(u_{i-1,J},v_{I-1,j},u_{i,J},v_{I,j},v_{I-1,j+1},v_{I,j+1},u_{i+1,J}\right)=\\
&a_{x,W}\left(u_{i-1,J},u_{i,J}\right)+a_{x,S}\left(v_{I-1,j},v_{I,j}\right)+a_{x,N}\left(v_{I-1,j+1},v_{I,j+1}\right)+a_{x,E}\left(u_{i,J},u_{i+1,J}\right)+\cdots\\
&\cdots+F_{x,e}\left(u_{i,J},u_{i+1,J}\right)-F_{x,w}\left(u_{i-1,J},u_{i,J}\right)+F_{x,n}\left(v_{I-1,j+1},v_{I,j+1}\right)-F_{x,s}\left(v_{I-1,j},v_{I,j}\right),
\end{aligned}\tag{6}$$

$$\begin{aligned}
&a_{y,W}\left(u_{i,J},u_{i,J-1}\right)=\max\left(F_{y,w}\left(u_{i,J},u_{i,J-1}\right),D_{y,w}+\frac{1}{2}F_{y,w}\left(u_{i,J},u_{i,J-1}\right),0\right),\\
&a_{y,S}\left(v_{I,j},v_{I,j+1}\right)=\max\left(F_{y,s}\left(v_{I,j},v_{I,j+1}\right),D_{y,s}+\frac{1}{2}F_{y,s}\left(v_{I,j},v_{I,j+1}\right),0\right),\\
&a_{y,N}\left(v_{I,j-1},v_{I,j}\right)=\max\left(-F_{y,n}\left(v_{I,j-1},v_{I,j}\right),D_{y,n}-\frac{1}{2}F_{y,n}\left(v_{I,j-1},v_{I,j}\right),0\right),\\
&a_{y,E}\left(u_{i+1,J},u_{i+1,J-1}\right)=\max\left(-F_{y,e}\left(u_{i+1,J},u_{i+1,J-1}\right),D_{y,e}-\frac{1}{2}F_{y,e}\left(u_{i+1,J},u_{i+1,J-1}\right),0\right),
\end{aligned}\tag{7}$$

$$\begin{aligned}
&a_{y,P}\left(u_{i,J},u_{i,J-1},v_{I,j},v_{I,j+1},v_{I,j-1},u_{i+1,J},u_{i+1,J-1}\right)=\\
&a_{y,W}\left(u_{i,J},u_{i,J-1}\right)+a_{y,S}\left(v_{I,j},v_{I,j+1}\right)+a_{y,N}\left(v_{I,j-1},v_{I,j}\right)+a_{y,E}\left(u_{i+1,J},u_{i+1,J-1}\right)\\
&+F_{y,e}\left(u_{i+1,J},u_{i+1,J-1}\right)-F_{y,w}\left(u_{i,J},u_{i,J-1}\right)+F_{y,n}\left(v_{I,j-1},v_{I,j}\right)-F_{y,s}\left(v_{I,j},v_{I,j+1}\right),
\end{aligned}\tag{8}$$

with $D_{\bullet,\bullet}$ a constant being a function of the viscosity μ (assumed constant) and the local grid sizes.

2.1. Wind turbine source terms

We use a static wind turbine model, and assume that the wind turbines are oriented in the axial direction, which results in the following expression for the source terms for cells in which a wind turbine is located:

$$\begin{aligned} &S_{x,i,J}V_{i,J} = -\frac{1}{2}\rho U_\infty^2 C_T\left(\beta,\lambda\right)\left(y_{i,j+1}-y_{i,j}\right) \text{ with } \lambda = \frac{\omega R}{U_\infty}, \\ &S_{y,I,j}V_{I,j} = 0, \end{aligned} \tag{9}$$

where U_∞ is the effective wind speed. This effective wind speed is found from the velocity profile around the turbine in the way described in [2]. The variable C_T represents the thrust coefficient of the turbine, which is a known function of the pitch angle β and the tip speed ratio λ, which in turn is a function of the rotor speed ω, the rotor radius R and the effective wind speed U_∞. In the simulation example of section 7, pitch β and rotor speed ω are chosen as a function of U_∞ according to a certain control policy optimizing the power production of the individual wind turbine.

2.2. Boundary conditions

Let N_x denote the number of grid cells in the x-direction, and N_y the number of grid cells in the y-direction. At the inflow boundaries of the control volume (South for v and West for u), we prescribe the incoming wind field by using:

$$\begin{aligned} &\text{West}: \; a_x\left(\mathbf{u}\right) = -a_{x,P}u_{i,J} + a_{x,W}u^{in}_{i-1,J} + a_{x,E}u_{i+1,J} + a_{x,N}u_{i,J+1} + a_{x,S}u_{i,J-1} \\ &\qquad \text{for } i = 3, J = 2,\cdots,N_y-1, \\ &\text{South}: \; a_y\left(\mathbf{u}\right) = -a_{y,P}v_{I,j} + a_{y,W}v_{I-1,j} + a_{x,E}v_{I+1,j} + a_{y,N}v_{I,j+1} + a_{y,S}v^{in}_{I,j-1} \\ &\qquad \text{for } I = 2,\cdots,N_x-1, \; j = 3, \end{aligned} \tag{10}$$

where u^{in}, v^{in} are the inflow boundary velocities assumed to be known. Similarly, by substituting the known inflows in (1b) we can obtain the continuity equations for the inflow boundaries. At the outflow boundaries of the control volume, we define a zero normal gradient boundary condition, by setting:

$$\begin{aligned} &\text{East:} \quad u_{N_x,J} = u_{N_x-1,J} \text{ for } J = 2,\cdots,N_y-2, \quad \text{and} \quad v_{N_x,j} = v_{N_x-1,j} \text{ for } j = 3,\cdots,N_y-2, \\ &\text{North:} \quad u_{i,N_y} = u_{i,N_y-1} \text{ for } i = 3,\cdots,N_x-2, \quad \text{and} \quad v_{I,N_y} = v_{I,N_y-1} \text{ for } I = 2,\cdots,N_x-1, \\ &\text{South:} \quad u_{i,1} = u_{i,2} \text{ for } i = 3,\cdots,N_x-1, \\ &\text{West:} \quad v_{1,j} = v_{2,j} \text{ for } j = 3,\cdots,N_y-1. \end{aligned} \tag{11}$$

Substituting these relations in the momentum equations (1a), yields:

$$
\begin{aligned}
&\text{East}: a_x(\mathbf{u}) = (-a_{x,P} + a_{x,E})\, u_{i,J} + a_{x,W} u_{i-1,J} + a_{x,N} u_{i,J+1} + a_{x,S} u_{i,J-1} \\
&\qquad \text{for } i = N_x - 1,\ J = 2, \cdots, N_y - 2, \\
&\text{North}: a_x(\mathbf{u}) = (-a_{x,P} + a_{x,N})\, u_{i,J} + a_{x,W} u_{i-1,J} + a_{x,E} u_{i+1,J} + a_{x,S} u_{i,J-1} \\
&\qquad \text{for } i = 3, \cdots, N_x - 2,\ J = N_y - 1, \\
&\text{South}: a_x(\mathbf{u}) = (-a_{x,P} + a_{x,S})\, u_{i,J} + a_{x,W} u_{i,J} + a_{x,E} u_{i+1,J} + a_{x,N} u_{i,J+1} \\
&\qquad \text{for } i = 3, \cdots, N_x - 1,\ J = 2, \\
&\text{East}: a_y(\mathbf{u}) = (-a_{y,P} + a_{x,E})\, v_{I,j} + a_{y,W} v_{I-1,j} + a_{y,N} v_{I,j+1} + a_{y,S} v_{I,j-1} \\
&\qquad \text{for } I = N_x - 1,\ j = 3, \cdots, N_y - 1, \\
&\text{North}: a_y(\mathbf{u}) = (-a_{y,P} + a_{y,N})\, v_{I,j} + a_{y,W} v_{I-1,j} + a_{x,E} v_{I+1,j} + a_{y,S} v_{I,j-1} \\
&\qquad \text{for } I = 2, \cdots, N_x - 1,\ j = N_y - 1, \\
&\text{West}: a_y(\mathbf{u}) = (-a_{y,P} + a_{y,W})\, v_{I,j} + a_{x,E} v_{I+1,j} + a_{y,N} v_{I,j+1} + a_{y,S} v_{I,j-1} \\
&\qquad \text{for } I = 2,\ j = 3, \cdots N_y - 1.
\end{aligned}
\tag{12}
$$

Similarly, the relations (11) can be substituted in (1b) to obtain the continuity equations for the East and North outflow boundaries.

3. Solution method for the nonlinear Navier-Stokes equations

To be able to find the time-variation of velocity field $\mathbf{u}(t)$, we apply backwards differencing in the time-domain to (1a), to obtain the fully discretized momentum equations:

$$
\begin{bmatrix} \frac{\rho V_{i,J}}{\Delta t} u_{i,J}^{k-1} + S_{x,i,J}^{k} V_{i,J} \\ \frac{\rho V_{I,j}}{\Delta t} v_{I,j}^{k-1} + S_{y,I,j}^{k} V_{I,j} \end{bmatrix} = \begin{bmatrix} a_x\left(\mathbf{u}^{k-1}, \mathbf{u}^{k}\right) + \frac{\rho V_{i,J}}{\Delta t} u_{i,J}^{k} + a_{xc} p_{I,J}^{k} - a_{xc} p_{I-1,J}^{k} \\ a_y\left(\mathbf{u}^{k-1}, \mathbf{u}^{k}\right) + \frac{\rho V_{I,j}}{\Delta t} v_{I,j}^{k} + a_{yc} p_{I,J}^{k} - a_{yc} p_{I,J-1}^{k} \end{bmatrix} \tag{13}
$$

where k denotes the discrete time index. Initially, the coefficients $a_{\bullet,\bullet}$ in a_x, a_y are calculated from velocities in the preceding time step $\mathbf{u}^{k-1}$, while the arguments, being the velocities in the current time step $\mathbf{u}^{k}$, are to be solved through solving the linearized equations.

We can combine inner cell equations (13),(1b) with boundary cell equations (10),(12) and the continuity equations for the boundaries, to obtain a system of linear equations. By solving this system we obtain:

$$
\begin{aligned}
&u_{i,J}^{k} \text{ for } i = 3, \cdots, N_x - 1,\ J = 2, \cdots, N_y - 1, \\
&v_{I,j}^{k} \text{ for } I = 2, \cdots, N_x - 1,\ j = 3, \cdots, N_y - 1, \\
&p_{I,J}^{k} \text{ for } I = 2, \cdots, N_x - 1,\ J = 2, \cdots, N_y - 1.
\end{aligned}
\tag{14}
$$

The coefficients $a_{\bullet,\bullet}$ in a_x, a_y can then be calculated from the above solution. After these updates we can solve the system again. We iterate several times in this way in each time step to obtain a more accurate solution of the nonlinear equations, and then proceed to the next time step. The MATLAB program implementing these different steps can be

found in the WindFarmSimulator package under the name `simWP.m`. In Appendix A, some plots can be found of the steady-state flows in wind parks with different configurations, as calculated in the manner described above.
Note that the above solution method may be used in some Model Predictive Control (MPC) approaches to do input-output predictions, or in model-free control approaches to verify the results, but to obtain a model for H_2 or H_∞ control synthesis, we would need a linearized control-oriented model, which describes the input-output relations between control and disturbance inputs, and performance and measurable outputs. To obtain this model, we need a linearization and projection step to be applied on (1). These steps are presented in the next sections.

4. Linearization

Let $\Delta x = x - x_{ss}$ denote the deviation of a variable x from a steady-state x_{ss}, then the linearized form of the Navier-Stokes momentum equations (1a) around a steady-state velocity field $\mathbf{u}_{ss}$ is given by:

$$\frac{\partial}{\partial t}\begin{bmatrix}\Delta u_{i,J}\\ \Delta v_{I,j}\end{bmatrix} = \begin{bmatrix}\frac{1}{\rho V_{i,J}} & 0\\ 0 & \frac{1}{\rho V_{I,j}}\end{bmatrix}\left(\begin{bmatrix}\frac{\partial a_x(\mathbf{u})}{\partial \mathbf{u}}\\ \frac{\partial a_v(\mathbf{u})}{\partial \mathbf{u}}\end{bmatrix}\Delta\mathbf{u} + \begin{bmatrix}a_{xc} & -a_{xc} & 0\\ 0 & -a_{yc} & a_{yc}\end{bmatrix}\begin{bmatrix}\Delta p_{I-1,J}\\ \Delta p_{I,J}\\ \Delta p_{I,J-1}\end{bmatrix} + \begin{bmatrix}\Delta S_{x,i,J}V_{i,J}\\ \Delta S_{y,I,j}V_{I,j}\end{bmatrix}\right), \tag{15}$$

with:

$$\begin{bmatrix}\frac{\partial a_x(\mathbf{u})}{\partial \mathbf{u}}\\ \frac{\partial a_v(\mathbf{u})}{\partial \mathbf{u}}\end{bmatrix}\Delta\mathbf{u} = \begin{bmatrix}\partial a_x/\partial u_{i-1,J} & \partial a_y/\partial u_{i-1,J}\\ \partial a_x/\partial v_{I-1,j} & \partial a_y/\partial v_{I-1,j}\\ \partial a_x/\partial u_{i,J-1} & \partial a_y/\partial u_{i,J-1}\\ \partial a_x/\partial v_{I,j-1} & \partial a_y/\partial v_{I,j-1}\\ \partial a_x/\partial u_{i,J} & \partial a_y/\partial u_{i,J}\\ \partial a_x/\partial v_{I,j} & \partial a_y/\partial v_{I,j}\\ \partial a_x/\partial u_{i,J+1} & \partial a_y/\partial u_{i,J+1}\\ \partial a_x/\partial v_{I,j+1} & \partial a_y/\partial v_{I,j+1}\\ \partial a_x/\partial u_{i+1,J} & \partial a_y/\partial u_{i+1,J}\\ \partial a_x/\partial v_{I+1,j} & \partial a_y/\partial v_{I+1,j}\end{bmatrix}^T \begin{bmatrix}\Delta u_{i-1,J}\\ \Delta v_{I-1,j}\\ \Delta u_{i,J-1}\\ \Delta v_{I,j-1}\\ \Delta u_{i,J}\\ \Delta v_{I,j}\\ \Delta u_{i,J+1}\\ \Delta v_{I,j+1}\\ \Delta u_{i+1,J}\\ \Delta v_{I+1,j}\end{bmatrix}, \tag{16}$$

where:

$$\begin{aligned}
\frac{\partial a_x}{\partial u_{i-1,J}} &= a_{x,W} + \frac{\partial a_{x,W}}{\partial u_{i-1,J}}u_{i-1,J} - \frac{\partial a_{x,P}}{\partial u_{i-1,J}}u_{i,J},\\
\frac{\partial a_x}{\partial v_{I-1,j}} &= \frac{\partial a_{x,S}}{\partial v_{I-1,j}}u_{i,J-1} - \frac{\partial a_{x,P}}{\partial v_{I-1,j}}u_{i,J},\\
\frac{\partial a_x}{\partial u_{i,J-1}} &= a_{x,S},\\
\frac{\partial a_x}{\partial v_{I,j-1}} &= 0,\\
\frac{\partial a_x}{\partial u_{i,J}} &= -a_{x,P} + \frac{\partial a_{x,W}}{\partial u_{i,J}}u_{i-1,J} + \frac{\partial a_{x,E}}{\partial u_{i,J}}u_{i+1,J} - \frac{\partial a_{x,P}}{\partial u_{i,J}}u_{i,J},\\
\frac{\partial a_x}{\partial v_{I,j}} &= \frac{\partial a_{x,S}}{\partial v_{I,j}}u_{i,J-1} - \frac{\partial a_{x,P}}{\partial v_{I,j}}u_{i,J},\\
\frac{\partial a_x}{\partial u_{i,J+1}} &= a_{x,N},\\
\frac{\partial a_x}{\partial v_{I,j+1}} &= \frac{\partial a_{x,N}}{\partial v_{I,j+1}}u_{i,J+1} - \frac{\partial a_{x,P}}{\partial v_{I,j+1}}u_{i,J},\\
\frac{\partial a_x}{\partial u_{i+1,J}} &= a_{x,E} + \frac{\partial a_{x,E}}{\partial u_{i+1,J}}u_{i+1,J} - \frac{\partial a_{x,P}}{\partial u_{i+1,J}}u_{i,J},\\
\frac{\partial a_x}{\partial v_{I+1,j}} &= 0,
\end{aligned}
\qquad
\begin{aligned}
\frac{\partial a_y}{\partial u_{i-1,J}} &= 0,\\
\frac{\partial a_y}{\partial v_{I-1,j}} &= a_{y,W},\\
\frac{\partial a_y}{\partial u_{i,J-1}} &= \frac{\partial a_{y,W}}{\partial u_{i,J-1}}v_{I-1,j} - \frac{\partial a_{y,P}}{\partial u_{i,J-1}}v_{I,j},\\
\frac{\partial a_y}{\partial v_{I,j-1}} &= a_{y,S} + \frac{\partial a_{y,N}}{\partial v_{I,j-1}}v_{I,j+1} - \frac{\partial a_{y,P}}{\partial v_{I,j-1}}v_{I,j},\\
\frac{\partial a_y}{\partial u_{i,J}} &= \frac{\partial a_{y,W}}{\partial u_{i,J}}v_{I-1,j} - \frac{\partial a_{y,P}}{\partial u_{i,J}}u_{i,J},\\
\frac{\partial a_y}{\partial v_{I,j}} &= -a_{y,P} - \frac{\partial a_{y,P}}{\partial v_{I,j}}v_{I,j} + \frac{\partial a_{y,S}}{\partial v_{I,j}}v_{I,j-1} + \frac{\partial a_{y,N}}{\partial v_{I,j}}v_{I,j+1},\\
\frac{\partial a_y}{\partial u_{i,J+1}} &= 0,\\
\frac{\partial a_y}{\partial v_{I,j+1}} &= a_{y,N} - \frac{\partial a_{y,P}}{\partial v_{I,j+1}}v_{I,j} + \frac{\partial a_{y,S}}{\partial v_{I,j+1}}v_{I,j-1},\\
\frac{\partial a_y}{\partial u_{i+1,J}} &= \frac{\partial a_{y,E}}{\partial u_{i+1,J}}v_{I+1,j} - \frac{\partial a_{y,P}}{\partial u_{i+1,J}}v_{I,j},\\
\frac{\partial a_y}{\partial v_{I+1,j}} &= a_{y,E}.
\end{aligned}$$

As the coefficients $a_{\bullet,\bullet}$ are piecewise linear in the velocities u, v, the derivatives $\frac{\partial a_{\bullet,\bullet}}{\partial u_{\bullet,\bullet}}$, $\frac{\partial a_{\bullet,\bullet}}{\partial v_{\bullet,\bullet}}$ are piecewise constant functions, easily found from the relations (3)-(8).

5. SSS/MSSS state-space structure

Note that the linearized system for each row of cells denoted by indices i and I, can be written as a distributed system with subsystems Σ_s given by:

$$\begin{bmatrix} E_s\dot{x}_s \\ \phi^p_{s-1} \\ \phi^m_{s+1} \\ z_s \\ y_s \end{bmatrix} = \begin{bmatrix} A_s & B^p_s & B^m_s & B^1_s & B^2_s \\ C^p_s & W^p_s & 0 & L^p_s & V^p_s \\ C^m_s & 0 & W^m_s & L^m_s & V^m_s \\ C^1_s & J^p_s & J^m_s & D^{11}_s & D^{12}_s \\ C^2_s & H^p_s & H^m_s & D^{21}_s & D^{22}_s \end{bmatrix} \begin{bmatrix} x_s \\ \phi^p_{s-1} \\ \phi^m_{s-1} \\ w_s \\ p_s \end{bmatrix} \tag{17}$$

for each $s = 1, \ldots, N_x-2$. For each row of cells denoted by the indices $i = s+1$, $I = s+1$, the state vector x_s contains the velocities $\{\Delta u_{i,J}\}_{J=2}^{N_y-1}$ $\{\Delta v_{I,j}\}_{j=3}^{N_y-1}$ and p_s contains the pressures $\{\Delta p_{I,J}\}_{J=2}^{N_y-1}$. The variables ϕ^m_s, ϕ^p_s pass on the states and the pressures from neighboring systems, and w_s contains inflow speeds at the inflow boundary cells u^{in} and v^{in}, and the source inputs ΔS_x. While p_s has an equal size for each subsystem, the dimensions of the x_s and w_s variables at the boundaries of the grids, or at the cells corresponding to the source inputs (i.e. the wind turbine locations), are different from those for the other cells. Vector z_s gives the divergence of the velocities for each cell, i.e. C^1_s contains the coefficients $a_{x,c}, a_{y,c}$. Direct-feedthrough matrix D^{11}_s contains the same coefficients at the boundaries and $D^{12}_s = 0$, and it should be that $z_s = 0$ to fulfill the divergence-free constraint. The vector y_s denotes the observed output, it is preferred not to be a function of p_s because this would yield some difficulties in the projection described in the next session, i.e. $D^{22}_s = 0$. Matrix E_s is a positive definite diagonal matrix containing the coefficients $\rho V_{i,J}$.
Together, the 1-D string of local systems Σ_s form a global interconnected system that has the following descriptor state-space form with an additional constraint:

$$\bar{E}\frac{d}{dt}\bar{x}(t) = \bar{A}\bar{x}(t) + \bar{B}_1\bar{w}(t) + \bar{B}_2\bar{p}(t), \tag{18a}$$

$$0 = \bar{C}_1\bar{x}(t) + \bar{D}_{11}\bar{w}(t), \tag{18b}$$

$$\bar{x}(0) = \bar{x}_0, \tag{18c}$$

$$\bar{y}(t) = \bar{C}_2\bar{x}(t) + \bar{D}_{21}\bar{w}(t), \tag{18d}$$

with:

$$\bar{C}_1 = \bar{B}_2^T, \tag{19}$$

and $\bar{x} \in \mathbb{R}^n$, $\bar{p} \in \mathbb{R}^{r_p}$, $\bar{w} \in \mathbb{R}^{r_w}$ and $\bar{y} \in \mathbb{R}^l$. The matrices $\bar{E}$, $\bar{A}$, $\bar{B}_1$, $\bar{B}_2$, $\bar{C}_1$, $\bar{C}_2$, $\bar{D}_{11}$, $\bar{C}_2$, $\bar{D}_{21}$ have the Sequentially Semi-Separable (SSS) structure, see [4], Chapter 4.
In a similar manner as described above, we can define a distributed system with local subsystems for each cell, and interconnect them as a state-space system with Multi-level

Sequentially Semi-Separable (MSSS) matrices, see Chapter 8 in [4].
Note that because $\bar{E}$ is a positive definite diagonal matrix, the above descriptor form can easily be converted to a regular state-space form, but the above form, with $\bar{B}_2$ appearing twice in the equations, is convenient for the Helmoltz-Leray projection, presented in the following section.

6. Helmholtz-Leray projection

As noted by [6]:

> "A fundamental observation associated with [[the Navier-Stokes equations (1)]] is that the momentum equations (1a) should be viewed as the equations that describe the velocity field, while the divergence-free condition (1b) provides an equation for pressure. [[...]] Indeed, it is often said that pressure enforces the divergence-free constraint in an incompressible flow. But this equation does not explicitly contain the pressure, and this has caused significant problems from a computational standpoint [[...]]."

In [5] a projection of the linearized Navier-Stokes equations (18) is presented that enforces the divergence-free constraint (18b) and eliminates $\bar{u}$ in the linearized momentum equations (18a), transforming the system in an Ordinary Differential Equation (ODE). The full derivation of this so-called Helmholtz-Leray type of projection is presented in the following [1]. This derivation is started with the observation that:

$$\bar{x}\left(t\right) = \tilde{x}\left(t\right) - \bar{E}^{-1}\bar{B}_2\left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{D}_{11}\bar{w}\left(t\right) \tag{20}$$

is a solution to (18b) if $\tilde{x}$ satisfies $0 = \bar{B}_2^T\tilde{x}\left(t\right)$. If we insert this solution into (18a) this yields:

$$\bar{E}\frac{d}{dt}\tilde{x}\left(t\right) = \bar{A}\tilde{x}\left(t\right) + \bar{B}_2\bar{p}\left(t\right) + \tilde{B}\bar{w}\left(t\right) - \bar{B}_2\left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{D}_{11}\frac{d\bar{w}\left(t\right)}{dt}, \tag{21a}$$

$$0 = \bar{B}_2^T\tilde{x}\left(t\right), \tag{21b}$$

$$\tilde{x}\left(0\right) = \bar{x}_0 - \tilde{D}\bar{w}\left(0\right), \tag{21c}$$

$$\bar{y}\left(t\right) = \bar{C}_2\tilde{x}\left(t\right) + \left(\bar{D}_{21} - \bar{C}_2\tilde{D}\right)\bar{w}\left(t\right), \tag{21d}$$

with $\tilde{D} = \bar{E}^{-1}\bar{B}_2\left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{D}_{11}$ and $\tilde{B} = \bar{B}_1 - \bar{A}\tilde{D}$. If we left-multiply the state-equation (21a) with $\bar{B}_2^T\bar{E}^{-1}$, use the fact that (21b) implies that $\bar{B}_2^T\frac{d}{dt}\tilde{x}\left(t\right) = 0$, and solve for $\bar{p}$, we obtain:

$$\bar{p}\left(t\right) = -\left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{B}_2^T\bar{E}^{-1}\bar{A}\tilde{x}\left(t\right) - \left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{B}_2^T\bar{E}^{-1}\tilde{B}\bar{w}\left(t\right) + \cdots \tag{22}$$

$$\cdots + \left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{D}_{11}\frac{d\bar{w}\left(t\right)}{dt}. \tag{23}$$

[1] as I think there is an error in the projection presented in [5], section 6

Inserting this in the state equation (21a) eliminates the derivative $\frac{d\bar{w}(t)}{dt}$:

$$\bar{E}\frac{d}{dt}\tilde{x}\left(t\right) = \Pi\bar{A}\tilde{x}\left(t\right) + \Pi\tilde{B}\bar{w}\left(t\right), \tag{24}$$

with $\Pi = I - \bar{B}_2\left(\bar{B}_2^T\bar{E}^{-1}\bar{B}_2\right)^{-1}\bar{B}_2^T\bar{E}^{-1}$.
Π is an oblique projector with $\text{null}\left(\Pi\right) = \text{range}\left(\bar{B}_2\right)$ and $\text{range}\left(\Pi\right) = \text{null}\left(\bar{B}_2^T\bar{E}^{-1}\right)$, which implies the following property of a vector φ in $\text{range}\left(\Pi\right)$:

$$\bar{B}_2^T\bar{E}^{-1}\varphi = 0 \quad \Leftrightarrow \quad \Pi\varphi = \varphi.$$

Further, using the property that $\Pi\bar{E} = \bar{E}\Pi^T$, and the change of variable $\tilde{x} = \bar{E}^{-1}\varphi$, this yields:

$$\bar{B}_2^T\tilde{x} = 0 \quad \Leftrightarrow \quad \Pi^T\tilde{x} = \tilde{x}. \tag{25}$$

Hence, (24) has to be solved for $\Pi^T\bar{x}\left(t\right) = \bar{x}$. If we insert this identity and left-multiply the equation with Π, and use the property of a projector that $\Pi^2 = \Pi$, this yields:

$$\Pi\bar{E}\Pi^T\frac{d}{dt}\tilde{x}\left(t\right) = \Pi\bar{A}\Pi^T\tilde{x}\left(t\right) + \Pi\tilde{B}\bar{w}\left(t\right) \tag{26a}$$

$$\bar{y}\left(t\right) = \bar{C}_2\Pi^T\tilde{x}\left(t\right) + \left[\bar{D}_{21} - \bar{C}_2\tilde{D}\right]\bar{w}\left(t\right), \tag{26b}$$

$$\tilde{x}\left(0\right) = \bar{x}_0 - \tilde{D}\bar{w}\left(0\right). \tag{26c}$$

In the above descriptor form of order n, the matrix $\Pi\bar{E}\Pi^T$ is a singular matrix with a maximum rank of $n - r_p$. The system can be collapsed in a descriptor form of order $n - r_p$ by performing a rank-revealing decomposition on Π, e.g. a Singular Value Decomposition:

$$\Pi = U_1\Sigma{V_1}^T = \Theta_l\Theta_r^T,$$

with $\Theta_l = U_1\Sigma \in \mathbb{R}^{n\times\tilde{n}}$ and $\Theta_r = V_1 \in \mathbb{R}^{n\times\tilde{n}}$ with $\tilde{n} \leq n - r_p$. Since Π is a projector for which it holds that $\Pi^2 = \Pi$, it holds that $\Pi = \Theta_l^T\Theta_r = I$.

$$\Theta_r^T\bar{E}\Theta_r\frac{d}{dt}\hat{x}\left(t\right) = \Theta_r^T\bar{A}\Theta_r\hat{x}\left(t\right) + \Theta_r^T\tilde{B}\bar{w}\left(t\right) \tag{27a}$$

$$\bar{y}\left(t\right) = \bar{C}_2\Theta_r\hat{x}\left(t\right) + \left[\bar{D}_{21} - \bar{C}_2\tilde{D}\right]\bar{w}\left(t\right), \tag{27b}$$

$$\hat{x}\left(0\right) = \Theta_l^T\bar{x}_0 - \Theta_l^T\tilde{D}\bar{w}\left(0\right), \tag{27c}$$

with $\hat{x} = \Theta_l^T\tilde{x} \in \mathbb{R}^{\tilde{n}}$. Using the property that $\Theta_r^{-1} = \Theta_r^T$, and the fact that $\bar{E}$ is positive-definite and diagonal, we can easily invert the matrix $\Theta_r^T\bar{E}\Theta_r$ to obtain a regular state-space system:

$$\frac{d}{dt}\hat{x}\left(t\right) = \left[\Theta_r^T\bar{E}^{-1}\bar{A}\Theta_r\right]\hat{x}\left(t\right) + \left[\Theta_r^T\bar{E}^{-1}\tilde{B}\right]\bar{w}\left(t\right). \tag{28a}$$

$$\bar{y}\left(t\right) = \left[\bar{C}_2\Theta_r\right]\hat{x}\left(t\right) + \left[\bar{D}_{21} - \bar{C}_2\tilde{D}\right]\bar{w}\left(t\right). \tag{28b}$$

$$\hat{x}\left(0\right) = \Theta_l^T\bar{x}_0 - \Theta_l^T\tilde{D}\bar{w}\left(0\right). \tag{28c}$$

7. Simulation example

We verify the obtained model (28) by performing a simulation of the flow profile around a single wind turbine, using three methods:

1. CFD simulation of the discretized nonlinear Navier-Stokes model (1) using the iterative time integration methods as explained in Section 3.

2. Simulation of the linearized and discretized state-space model (18a),(18c),(18d) with the pressure profile $\bar{p}\,(t) = 0$, just to show that including the pressure profile, i.e. the divergence-constraint, has a significant effect.

3. Simulation of the linearized and discretized state-space model (18a),(18c),(18d) with the pressure profile $\bar{p}\,(t)$ as generated by the aforementioned CFD simulation.

4. Simulation of the discretized, linearized and projected model (28).

In the discretization, we use a small number of cells: $N_x = 25$ cells in the x-direction and $N_y = 10$ cells in the y-direction, to make the projection computationally feasible with the standard MATLAB linear algebra operations. In each case, we excite the systems for 200s with a small variation of the East incoming wind speed:

$$u^{in}_{i,J}\,(t) = 10 + 0.05\sin(2\pi t/50) + 0.05 u_{rand}\,(t)$$

for $i = 1$, $J = 1, \ldots, N_y$, where u_{rand} is unit-variance Gaussian noise. The South incoming wind field v^{in} is zero. This incoming wind speed profile determines, together with the forces S_x from the wind turbine, the disturbance vector $\bar{w}$ in the methods 2, 3 and 4. In each method, the source terms S_x are generated by an actuator disk model in the CFD simulation. In methods 2, 3 and 4, the operating point for linearization is the steady-state wind field for a constant incoming wind speed of 10m/s, and we note that the variation from this operating point is quite small. The steady-state also defines the initial state, i.e. $\bar{x}_0 = 0$. See Figure 2 for a plot of the flow fields around the wind turbine in steady state, as calculated with method 1.
From the wind-speed profiles generated by each method, we calculate the power produced by the wind turbines, which is a function of the effective wind speed in front of the turbine. We evaluate the effect of the linearization by calculating how well the power predicted by simulation methods 2 to 3 fit the power predicted by method 1, see Figure 3 and Table 1. We see that the projected system output (method 2) indeed closely fits the CFD solution, for these small excitations around the linearization point. An unexpected result is that the linearized model with the pressure field from the CFD model (method 3) does not perform that well.

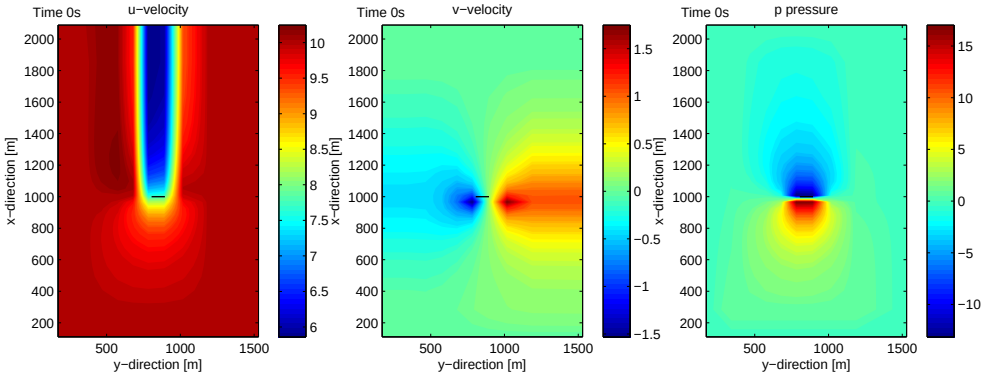


Figure 2: Plot of steady-state wind field around a single wind turbine, as calculated by the CFD solver described in section 3. Calculated with 20 iterations on a 25x10 meshing grid.

	VAF %	BFR %
method 2: lin. model no pressure	44.7	24.5
method 3: lin. model with $\bar{p}$ from CFD	85.8	54.8
method 4: projected lin. model	98.3	86.9

Table 1: VAF (Variance Accounted For) and BFT (Best Fit Ratio) of the predicted wind turbine produced power for simulation methods 2 to 4 compared to method 1.

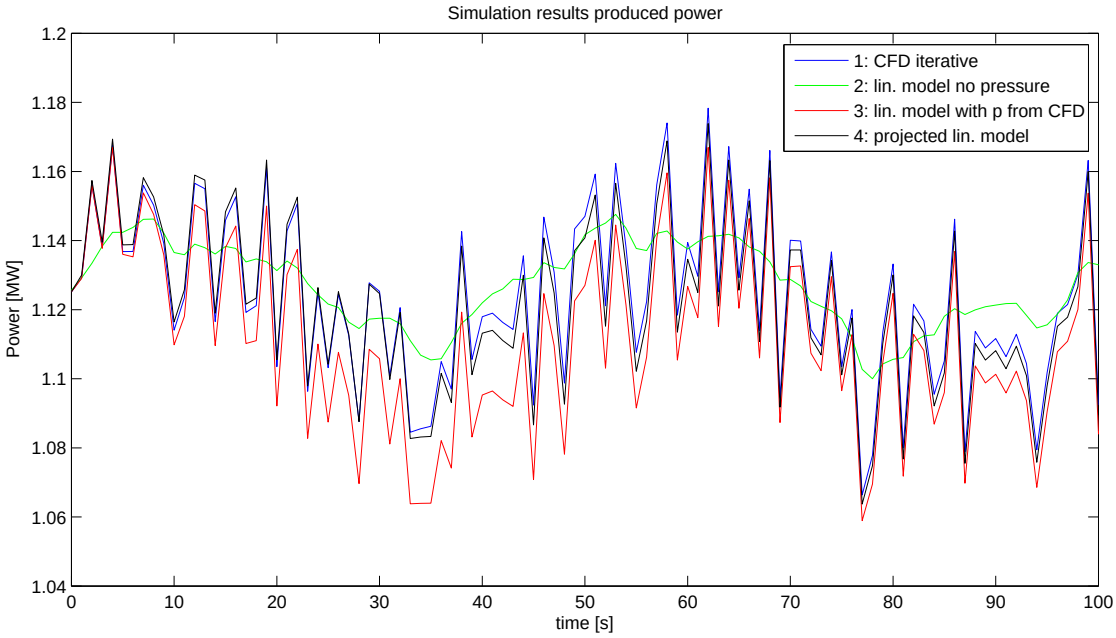


Figure 3: Predicted wind turbine produced power for simulation methods 1 to 4

8. Conclusions and future work

This report shows how to obtain a control-oriented linearized model from the Navier-Stokes equations that are used in the WindFarmSimulator to describe the flow through a wind park. The final model describes the input-output relation between variation of the incoming wind field at the boundaries of the volume under consideration, which we can see as disturbance inputs, the forces exerted by the wind turbines, over which we have some control, and the output in terms of the wind flow variations. The explicit dependence on the pressure is lost by projection of the equations on the divergence-free constraint.
The future work includes resolving the following issues:

- Although the focus of this research is not to produce a realistic wind park CFD simulation, but rather to test some control concepts in a fast simulation that may have some characteristics of the spatially distributed structure of the wind park dynamics, there is still a desire to make the fluid dynamics model more accurate, by resolving the following issues:
  - There is some doubt about how the boundary condition were set, and how they are used in solving the Navier-Stokes equations. In our method, we do not set additional boundary conditions on the pressure, and it is unclear to us whether this is needed.
  - An extension to a 3D simulation may be needed in order to include the effects of inflow of momentum form the upper and lower atmosphere, which may significantly increase the wake recovery. To keep the simulation time low, we would think of including a low number of grid cells in the z direction, maybe less than 10.
- We presented the analytical linearization of the nonlinear model. Of course, linearization makes that the model is only accurate for small deviations from the operating point, and we will have to use some scheduling to describe the model over the full operating range, perhaps in the form of an LPV model.
- The projection step may become quite computationally expensive for larger models. Note that [5] presents a method for simultaneous projection *and* model reduction for the linearized Navier-Stokes equations without calculating the projection matrix Π or performing the SVD on Π. So far, the SSS structure of the matrices was not exploited yet by using the SSS/MSSS toolbox, so I expect that there is a lot to gain there in terms of a speed-up of the computations.

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[6] J. M. McDonough, *Lectures in Computational Fluid Dynamics of Incompressible Flow: Mathematics, Algorithms and Implementations*, 2007.

Appendix A. Plots of simulation results with different wind park configuration

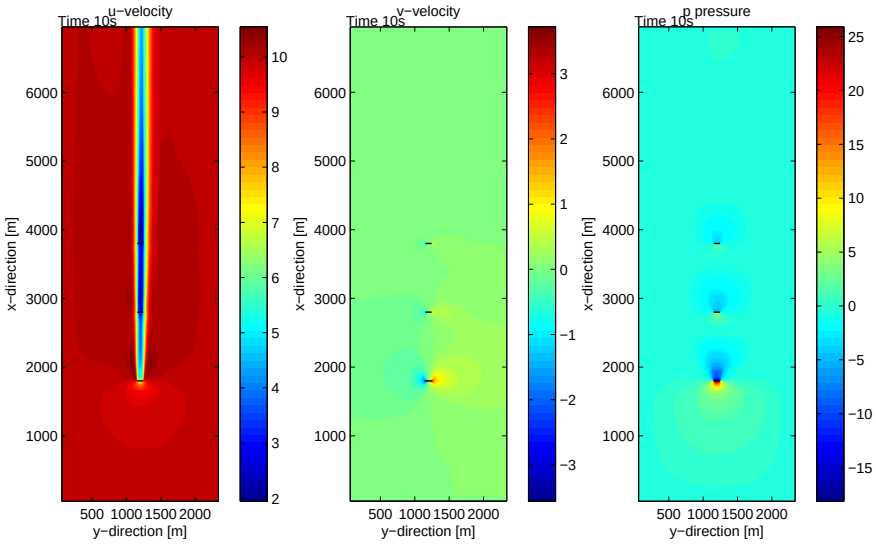


Figure 4: Plot of steady-state wind field with 3 wind turbines in each others wake, calculated by the CFD solver described in section 3. Calculated with 20 iterations on a 200x120 meshing grid.

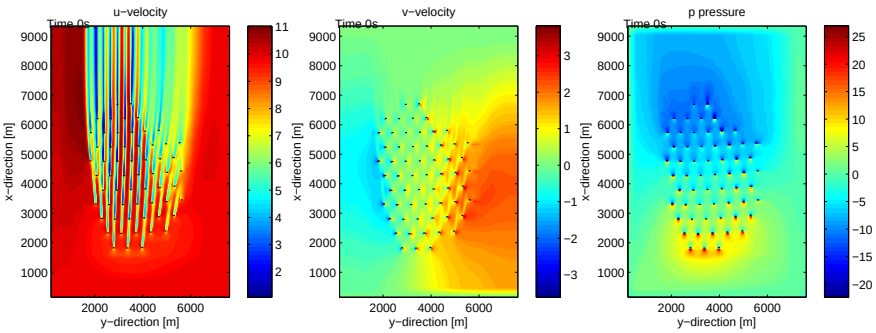


Figure 5: Plot of steady-state wind field in a wind park with the configuration of the Princess Amalia Wind Park, as calculated by the CFD solver described in section 3. Calculated with 20 iterations on a 420x350 meshing grid.